

TCS3351 Cryptography and Data Security
Tutorial 3
Classical Cryptosystems

1. Distinguish between the following:
 - i) Substitution and transposition cipher
 - ii) Monoalphabetic and polyalphabetic cipher

Answer:

- i) Substitution cipher replaces one symbol or character with another, while transposition cipher changes the location of the symbol or character.
 - ii) Monoalphabetic cipher is a one-to-one encipherment between a symbol in a plaintext to a symbol in the ciphertext. The symbol in a plaintext is always enciphered into the same symbol in the ciphertext regardless of its position. Polyalphabetic cipher is a one-to-many encipherment between a symbol in a plaintext to a symbol in the ciphertext. The symbol in a plaintext may be enciphered into different symbols in the ciphertext, depending on its position and the key stream used.
2. State the maximum number of characters that will be changed in the ciphertext if only a single character is changed in the plaintext, for each of the following ciphers. Briefly explain the reason for your answer.
 - i) Affine
 - ii) Auto-key
 - iii) One-time pad
 - iv) Rotor

Answer:

- i) 1. Affine cipher is a monoalphabetic cipher that encrypts character independently.
 - ii) 2. Since the plaintext will become part of the keystreams, when a character is changed, that ciphertext is changed, and the next ciphertext will be changed too. This character is the key for the next encipherment.
 - iii) 1. Each character is enciphered independently with a specific key in the keystream.
 - iv) 1. Each character is mapped to its corresponding ciphertext from the rotor.
3. State the two problems with one-time pad.

Answer:

- i) The key is as long as the plaintext, and only used once in each encipherment. A large quantity of keys is needed and hence impractical.
 - ii) The distribution of large quantities of key to all users makes the one-time pad inefficient and impractical.

4. Encrypt the message “exe” using the following ciphers. Show also how you can get the original message by performing decryption. In this

- i) Additive cipher with $Key = 20$.
- ii) Multiplicative cipher with $Key = 15$.
- iii) Affine cipher with $Key = (15, 20)$.

a	B	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

4(i) Additive Cipher Key = 20

X	E	X	E		
	4	23	4		
Key = 20	$4 + 20 = 24$	43 $= 43 \bmod 26$ $= 17$	24		
Cipher Text	Y (24)	$17 = R$	Y		
Key = -20	$24 - 20 = 4$	$17 - 20 = -3$ $-3 + 26 = 23$	$24 - 20 = 4$		
Plaintext	E	X = 23	E = 4		

4ii) Multiplicative Cipher Key = 15

X	E	X	E		
	4	23	5		
Key = 15	$4 \times 15 = 60$	$23 \times 15 = 345$	$4 \times 15 = 60$		
Cipher Text	$60 \bmod 26 = 8$ I	$345 \bmod 26 = 7$ H	I		
Decryption Key = 15^{-1} Is 7	$8 \times 7 = 56$	$7 \times 7 = 49$	$8 \times 7 = 56$		
Plaintext	$56 \bmod 26 = 4 = (E)$	$49 \bmod 26 = 23 (X)$	$56 \bmod 26 = 4 = (E)$		

4iii) Affine cipher $E(x) = 15x + 20$,

X	E	X	E		
	4	23	5		
Key1 = $15x$	60	345	60		
Key2 = +20	80	365	80		
Ciphertxt	$80 \bmod 26 = 2$	$365 \bmod 26 = 5$	2		

Decryption process

Ciphertext	$80 \bmod 26 = 2$	$365 \bmod 26 = 1$	2		
Key = $7y - 140$	$= 7(2) - 140 = -126$	$= 7(1) - 140 = -133$			
	$= [-126 + 5(26)] \bmod 26$	$= [-133 + 6(26)] \bmod 26 = 23 \bmod 26$			
Plaintext	$4 = E$	$23 = X$			

$$E(x) = y = 15x + 20$$

$$x = (y - 20) \cdot 15^{-1} ; 15^{-1} = 7.$$

$$x = 7(y - 20) = 7y - 140 \text{ *****}$$

$$D(x) = 7x - 140.$$

Affine cipher

$$E(x) = (ax + b) \bmod m$$

$$x = \{A=0, B=1, 2, \dots, Z=25\}$$

a, b are integers

Condition: 'a' must satisfy the following condition, $\gcd(a, m) = 1$;

To find inverse of a, a^{-1} .

$$(a^{-1}) \cdot a = 1 \bmod m.$$

$$A = 15. \quad 15x ? = L \bmod 26 = 1$$

$$15x \cdot 7 = 105 \bmod 26 = 1$$

Thus $a^{-1} = 7$

$$Y = (15x + 20) \bmod 26$$

$$\text{Let } y = 15x + 20$$

$$X = (y - 20) / 15.$$

$$= 15^{-1}(y - 20)$$

EEA.

$$15x + 26y = 1$$

$$26 = 15(1) + 11; \rightarrow 11 = 26 - 15 \dots\dots\dots \text{Eq 1}$$

$$15 = 11(1) + 4; \rightarrow 4 = 15 - 11 \dots\dots\dots \text{eq 2}$$

$$11 = 4(2) + 3; \rightarrow 3 = 11 - 4(2) \dots\dots\dots \text{eq 3}$$

$$4 = 3(1) + 1; \rightarrow 1 = 4 - 3 \dots\dots\dots \text{eq 4}$$

$$3 = 1(3) + 0$$

From eq 4.

$$1 = 4 - 3$$

$$1 = 4 - (11 - 4(2)) ; \quad \text{substitute eq 3 to 3.}$$

$$1 = 3(4) - 11 ;$$

$$1 = 3(15 - 11) - 11 ; \quad \text{substitute eq 2 to 4}$$

$$1 = 3(15) - 4(11)$$

$$1 = 3(15) - 4(26 - 15) ; \text{substitute eq 1 to 11}$$

$$1 = 7(15) - 4(26).$$

Thus, the inverse of 15 is 7.

$$15 \cdot 7 \bmod 26 = 105 \bmod 26 = 1.$$

Answer:

i) **YRY**

ii) **IHI**

iii) **RBR**

a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	U	v	w	x	y	Z
0	1	2	3	4	5	6	7	8	9	1	1	1	1	1	1	1	1	1	1	2	2	2	2	2	2
										0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5
-	-																			-	-	-	-	-	-
2	2																			6	5	4	3	2	1
5	4																								

5. Encrypt the following message using Playfair encryption technique. ***

“The key is hidden under the door pad”

Ignore the spaces between the words. Use **GUIDANCE** as the key. Show also how you can get the original message by performing decryption.

“The key is hidden under the door pad”

Answer:

G	U	I / J	D	A
N	C	E	B	F
H	K	L	M	O
P	Q	R	S	T
V	W	X	Y	Z

Ciphertext: **POCL BXDR LGIY(JY) IB(JB)CG BGLX POBI(BJ) LZLT TGIY(JY)**

- **Put in 2 character. Repeating letter pair. Pad with x. The key is hidden under the door pad”**
- **Th ek ey is hi dx de nu nd er th ed ox or pa dx**
- **If plaintext letter in same row, move right**
- **If plaintext letter in same column, move down**
- **If plaintext letter not in same row/column, swap column (most cases)**

A	B	C	D	E
F	G	H	I/J	K
L	M	N	O	P
Q	R	S	T	U
V	W	X	Y	Z

6. Encrypt the message “**meet me**” using the Hill cipher with the key $\begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$. Show also how you can get the original message by performing decryption. Hill Cipher key is 2x2, 3x3, 4x4 ... nxn, where the encryption and decryption process goes by n characters.

Answer:

$$ME: \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 12 \\ 4 \end{pmatrix} = \begin{pmatrix} 108 + 16 \\ 60 + 28 \end{pmatrix} = \begin{pmatrix} 124 \\ 88 \end{pmatrix} \equiv \begin{pmatrix} 20 \\ 10 \end{pmatrix} = \begin{pmatrix} U \\ K \end{pmatrix} \pmod{26}$$

$$ET: \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix} \begin{pmatrix} 4 \\ 19 \end{pmatrix} = \begin{pmatrix} 36 + 76 \\ 20 + 133 \end{pmatrix} = \begin{pmatrix} 112 \\ 153 \end{pmatrix} \equiv \begin{pmatrix} 8 \\ 23 \end{pmatrix} = \begin{pmatrix} I \\ X \end{pmatrix} \pmod{26}$$

Ciphertext: **UKIX UK** where c_1 & c_2 represent the cipher text.

Plain Text = AB , Cipher Text =

The inverse of key $\begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$ using Extended Euclidean Algorithm:

$$\begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}^{-1} = \frac{1}{43} \begin{pmatrix} 7 & -4 \\ -5 & 9 \end{pmatrix} \equiv \frac{1}{17} \begin{pmatrix} 7 & -4 \\ -5 & 9 \end{pmatrix} \pmod{26} = 23 \begin{pmatrix} 7 & -4 \\ -5 & 9 \end{pmatrix}$$

$$1/43 \pmod{26} = 1/(43-26) \pmod{26} = 1/17 \pmod{26}$$

$$17x \pmod{26} = 1 \pmod{26}$$

Try $x =$ from 1,... 25 until you get $17x \pmod{26} = 1$

It was found that $x = 23$, $17 \times 23 = 391$, $391 \pmod{26} = 1$

$$391 = 15 \times 26 + 1$$

Use EEA,

$$17x + 26y = 1$$

EEA

$$26 = 17(1) + 9$$

$$17 = 9(1) + 8$$

$$9 = 8(1) + 1$$

$$8 = 1(8) + 0, \text{ GCD} = 1$$

Reverse the process, the key is put it in the form of $17x + 26y = 1$

$$1 = 9 - 8(1)$$

$$8 = 17 - 9(1)$$

$$9 = 26 - 17$$

$$1 = 9 - 8$$

$$= 9 - (17 - 9)$$

$$= 2(9) - 17$$

$$\text{But } 9 = 26 - 17$$

So

$$1 = 2(26 - 17) - 17$$

$$= 2(26) - 3(17)$$

$$17^{-1} = -3 = 23$$

$$9 \cdot (9^{-1}) \pmod{26} = 1$$

Find what is the value of (9^{-1}) .

$$(9x + 26y) \pmod{26} = 1$$

Try $x = 1, \dots, 25$.

EEA

$$9x + 26y = 1$$

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$8 = 1(8) + 0$$

Put it in the form of $9x+26y = 1$

$$\begin{aligned} 1 &= 9 - 8(1) \\ &= 9 - (26 - 9(2)) \\ &= 9(3) - 26 \end{aligned}$$

Thus, 9^{-1} is 3.

Proof,

$$9 \cdot 3 \bmod 26 = 27 \bmod 26 = 1$$

$$UK: \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}^{-1} \begin{pmatrix} 20 \\ 10 \end{pmatrix} \equiv_{26} \begin{pmatrix} 7 & -4 \\ -5 & 9 \end{pmatrix} \begin{pmatrix} 20 \\ 10 \end{pmatrix} = \begin{pmatrix} 12 \\ 4 \end{pmatrix} = \begin{pmatrix} M \\ E \end{pmatrix} \pmod{26}$$

$$IX: \begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}^{-1} \begin{pmatrix} 8 \\ 23 \end{pmatrix} \equiv_{26} \begin{pmatrix} 7 & -4 \\ -5 & 9 \end{pmatrix} \begin{pmatrix} 8 \\ 23 \end{pmatrix} = \begin{pmatrix} 4 \\ 19 \end{pmatrix} = \begin{pmatrix} E \\ T \end{pmatrix} \pmod{26}$$

where p_1 & p_2 represent the plain text.

Basic Explanation.

Given a Matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the determinant of this matrix $\det(A) = ad - bc$

To find the inverse of this matrix, follow the following steps

Create a new matrix,

$$Adj(A) = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

For example,

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

$$A^{-1} = \frac{1}{1(4) - 2(3)} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$$

$$A^{-1} = -\frac{1}{2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} -2 & 1 \\ 6 & -1/2 \end{pmatrix}$$

7. Encrypt the plaintext message “**the house is being sold tonight**” using the following ciphers. Ignore the space between words.

i) Vignere cipher with key: “**dollars**”

ii) Autokey cipher with $Key = 7$. =h

Show also how you can get the original message by performing decryption.

How Vignere Ciphers works?

Encryption rule: $C_i = (P_i + K_i) \bmod 26$

Decryption rule: $P_i = (C_i - K_i + 26) \bmod 26$

a	B	c	d	E	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
T	h	E	h	O	u	s	e	i	s	b	e	i	n	g	s	o	l	d	t	o	n	i	g	h	t
19	7	4	7	14	20	18		88	18	14	83	16	83	14	11	31	14		19	14	13	86	71	19	
d	o	L	l	A	r	s	d	o	l	l	a	r	s	d	o	L	l	a	r	s	d	o	l	l	A
3	14	11	11	0	17	18	34	11	11	10	17	18	34	11	11	11	0	17	18	34	11	11	11	0	
22	21	15	18																						
W	v	p	S																						

Auto key.

T	h	E	H	O	u	s	e	i	s	b	e	i	n	g	s	o	l	d	t	o	n	i	g	h	t
19	7	4	7	14	20	18	4	8	18	1	4	8	13	6	18	14	11	3	19	14	13	8	6	7	19
7	19	7	4	7	14	20	18	4	8	18	1	4	8	13	6	18	14	11	3	19	14	13	8	6	7
0	0	11	11	21																					
A	A	L	L	V																					

Answer:

i) WVPS OLKH WDME ZFJG ZWDK GQWR ST

ii) AALL VIMW MATF MVTY GZOWHBVO NA

Autokey:

Append the **plaintext**, **after the key** to create a key as long as the plaintext.

Example,

Plaintext : The house is being sold tonight

Key : hThe house is being sold tonigh

Then, the rest is same as vignere ciphers

Encryption rule: $C_i = (P_i + K_i) \bmod 26$

Decryption rule: $P_i = (C_i - K_i + 26) \bmod 26$

8. Identify the plaintext message using the **Rail Fence cipher of depth 2, 3 and 4**. Given ciphertext is

FLYILDODENTPL ULFLEPTOSOSIL

F		L		Y		I		L		D		O		D		E		N		T		P		L	
	U		L		F		L		E		P		T		O		S		O		S		I		L
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5

Answer:

Depth 2: **Fully Filled Pot Does Not Spill.**

Depth 3:

0				4				8				1				1				2				2	
	1		3		5		7		9		1		1		1		1		1		2		2		2
		2			6				1				1				1					2			
									0				4				8				2				

Rail 1: 7 characters

Rail 2: 13 characters

Rail3 : 6 characters

FLYILDODENTPLULFLEPTOSOSIL

FLY LID DOES NOT FILL TOO

Fully Filled Pot Does Not Spill.

F				Y				L				O													
	U		L		F		L		E		P														
		L				I				D															

Depth 4:

Rephrase Q: Plain Text : **Fully Filled Pot Does Not Spill.**

Rail fence Cipther Depth 4: FIONLUFLPTSOILYLDDDET PLEOSSL

[illegible]

i) First stage decryption: **2 4 6 5 1 3 7**

1	2	3	4	5	6	7
W	P	P	O	A	F	I
U	Y	S	A	I	H	T
E	R	T	F	U	I	H
A	S	S	L	S	Z	T
I	N	B	S	O	E	A
X	H	H	T	U	I	R
B	S	R	T	H	O	N
A	R	T	B	A	I	Z

2	4	6	5	1	3	7
P	O	F	A	W	P	I
Y	A	H	I	U	S	T
R	F	I	U	E	T	H
S	L	Z	S	A	S	T
N	S	E	O	I	B	A
H	T	I	U	X	H	R
S	T	O	H	B	R	N
R	B	I	A	A	T	Z

First output: **POFA WPIY AHIU STRF IUET HSLZ SAST NSEO IBAH TIUX HRST
OHBR NRBI AATZ**

Second-stage decryption: **2 4 6 5 1 3 7**

1	2	3	4	5	6	7
P	A	I	S	I	H	N
O	H	U	A	B	R	R
F	I	E	S	A	S	B
A	U	T	T	H	T	I
W	S	H	N	T	O	A
P	T	S	S	I	H	A
I	R	L	E	U	B	T
Y	F	Z	O	X	R	Z

2	4	6	5	1	3	7
A	S	H	I	P	I	N
H	A	R	B	O	U	R
I	S	S	A	F	E	B
U	T	T	H	A	T	I
S	N	O	T	W	H	A
T	S	H	I	P	S	A
R	E	B	U	I	L	T
F	O	R	X	Y	Z	Z

Plaintext: **A Ship In Harbour Is Safe But That Is Not What Ships Are Built For.**

10. In a **Polybius cipher**, each letter is enciphered as two integers. The key is a 5×5 matrix of characters as in a Playfair cipher. The plaintext is the character in the matrix, the ciphertext is the two integers (each between 1 and 5) representing row and column numbers. Encrypt the word “**multimedia**” using the Polybius cipher with the following key:

	1	2	3	4	5
1	z	q	P	f	e
2	y	r	O	g	d
3	x	s	N	h	c
4	w	t	M	i / j	b
5	v	u	L	k	a

Answer:

m	u	l	t	i	M	e	d	i	a
43	52	53	42	44	43	15	25	44	55

Ciphertext: **43 52 53 42 44 43 15 25 44 55**

References:

1. Behrouz A. Forouzan, Debdeep Mukhopadhyay, "Cryptography and Network Security", Third Edition, McGraw-Hill, 2015.
2. William Stallings, "Cryptography and Network Security – Principles and Practice", Seventh Edition, Prentice Hall, 2017.